

Supplementary Information

Activation entropy as a key factor controlling the memory effect in glasses

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1. Sample fabrication

The master alloy of $\text{Au}_{49}\text{Cu}_{26.9}\text{Ag}_{5.5}\text{Pd}_{2.3}\text{Si}_{16.3}$ (at.%) was first melt using arc melting furnace. Metallic glass ribbon was obtained by the melt spinning technique (tangent speed ~ 40 m/s). During melting, the alloy was protected by high-purity Ar. A tiny sample was cut from the ribbon under microscope and loaded on the flash DSC chip for measurement.

2. The DSC measurements

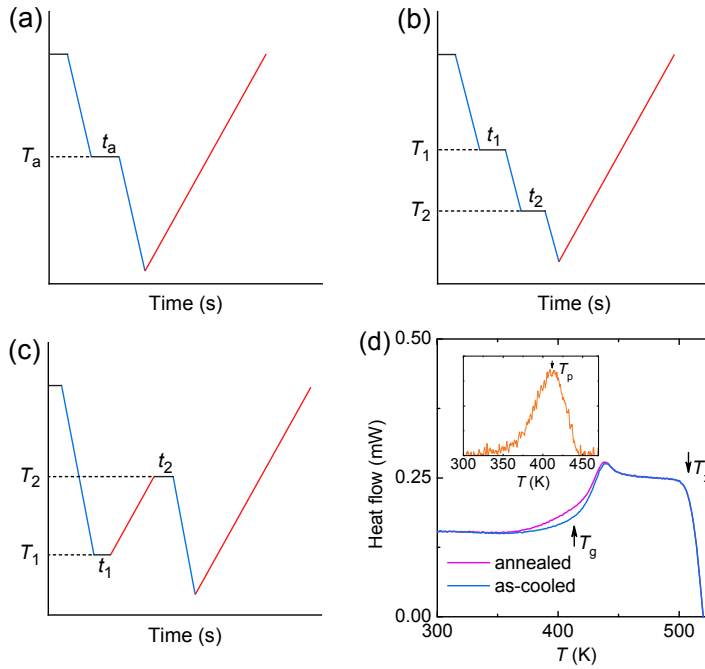


Figure S1. (a) Temperature-time protocols for single-step ageing. (b) Temperature-time protocols for two-step ageing. (c) Temperature-time protocols for memory effect. (d) The representative DSC curves obtained upon heating for as-cooled sample (blue) and the sample annealed at 383 K for 0.1 s (red). Inset is the relaxation peak obtained by subtracting the two DSC traces.

During measurement, a flow of high-purity Ar gas (40 ml/min) was applied to protect the sample from oxidation. The heat flow of the as-cooled sample was first measured at a heating rate of $R_h=1000$ K/s. The enthalpy change that occurred during isothermal annealing is given by $\Delta h = \int_{T_R}^{T_0} \Delta C_p dT$, with T_0 is the integral upper limit

(in the supercooled liquid state), T_R is room temperature (relaxation peak is above room temperature), and ΔC_p is the change in the specific heat. Schematic illustration of single and two-step thermal progresses are shown in Fig. S1(a)-(c). The DSC curves were measured at cooling rate $R_c=10000$ K/s and heating rate $R_h=1000$ K/s. As shown in Fig. S1(d), the glass transition temperature T_g and the onset crystallization temperature T_x are marked by arrows. Inset is the relaxation peak by subtracting the two DSC traces and the relaxation peak T_p is obtained.

3. 2-D contours of the relaxation heat flows for $T_1=383$ K and $T_2=373$ K

The relaxation heat flows evolve with the pre-ageing time $t_1=0.1$ -100 s for $T_1=383$ K and $T_2=373$ K experiments are shown in Figure S2. The results show that the relaxation heat flows monotonic increases with t_2 for the entire pre-ageing time range $t_1=0.1$ -100 s. The memory effect was not observed in this two step ageing experiments.

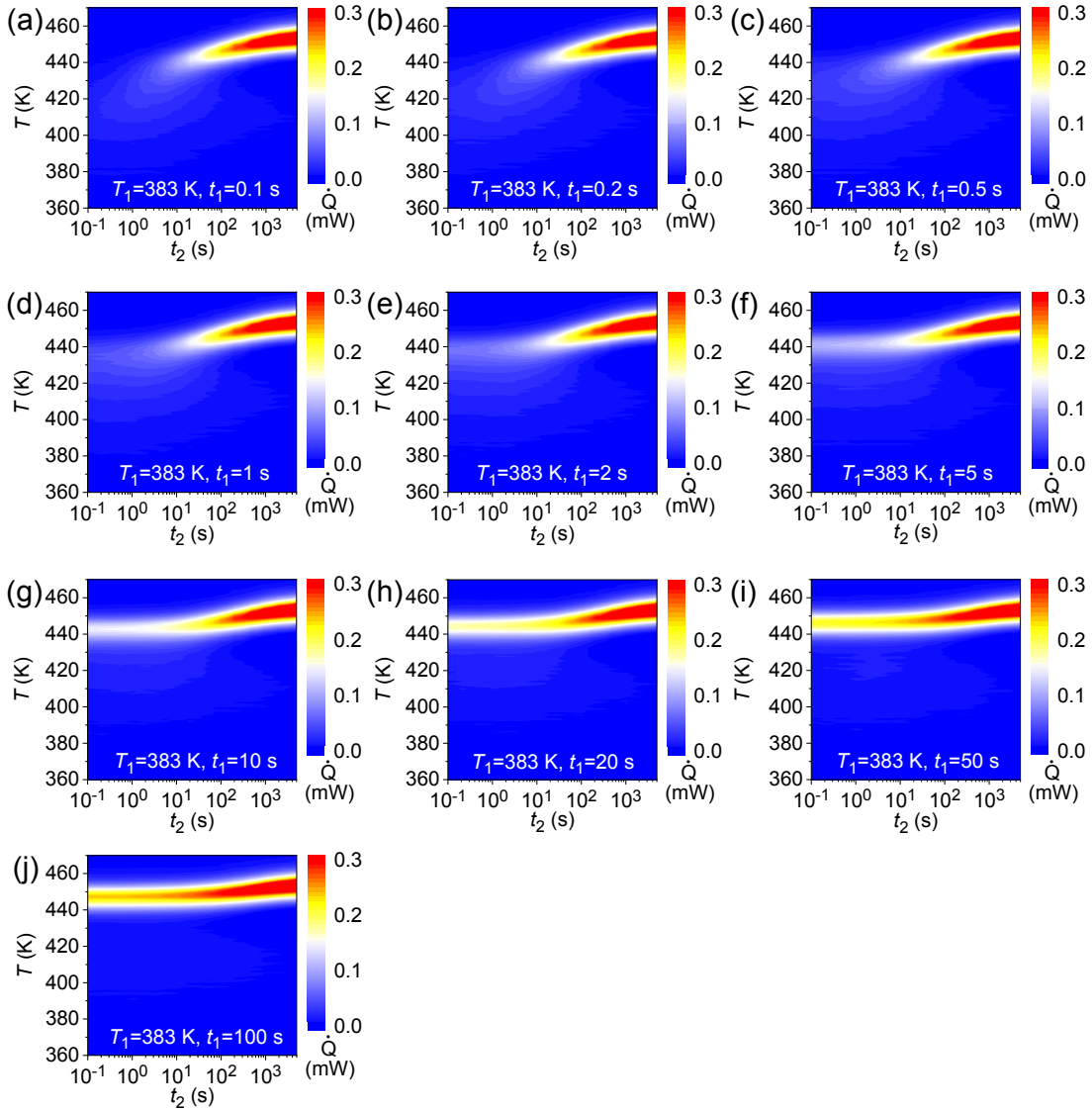


Figure S2. (a)-(j) Two-dimension contours of the relaxation heat flows of sample pre-ageing at $T_1=383$ K for $t_1=0.1$ -100 s and then ageing at $T_2=373$ K, $t_2=0.1$ -5000 s.

4. 2-D contours of the relaxation heat flows for memory effect ($T_1=373$ K and $T_2=383$ K)

To better illustrate the relaxation heat flows evolve with the pre-ageing time t_1 for memory effect, the two-dimensional contours of the relaxation heat flows pre-ageing at $T_1=373$ K for $t_1=0.1$ -100 s are shown in Figure S3. The results show that the relaxation heat flows monotonic increases with t_2 for $t_1=0.1$ -2 s but nonmonotonic changes with t_2 for $t_1=5$ -100 s.

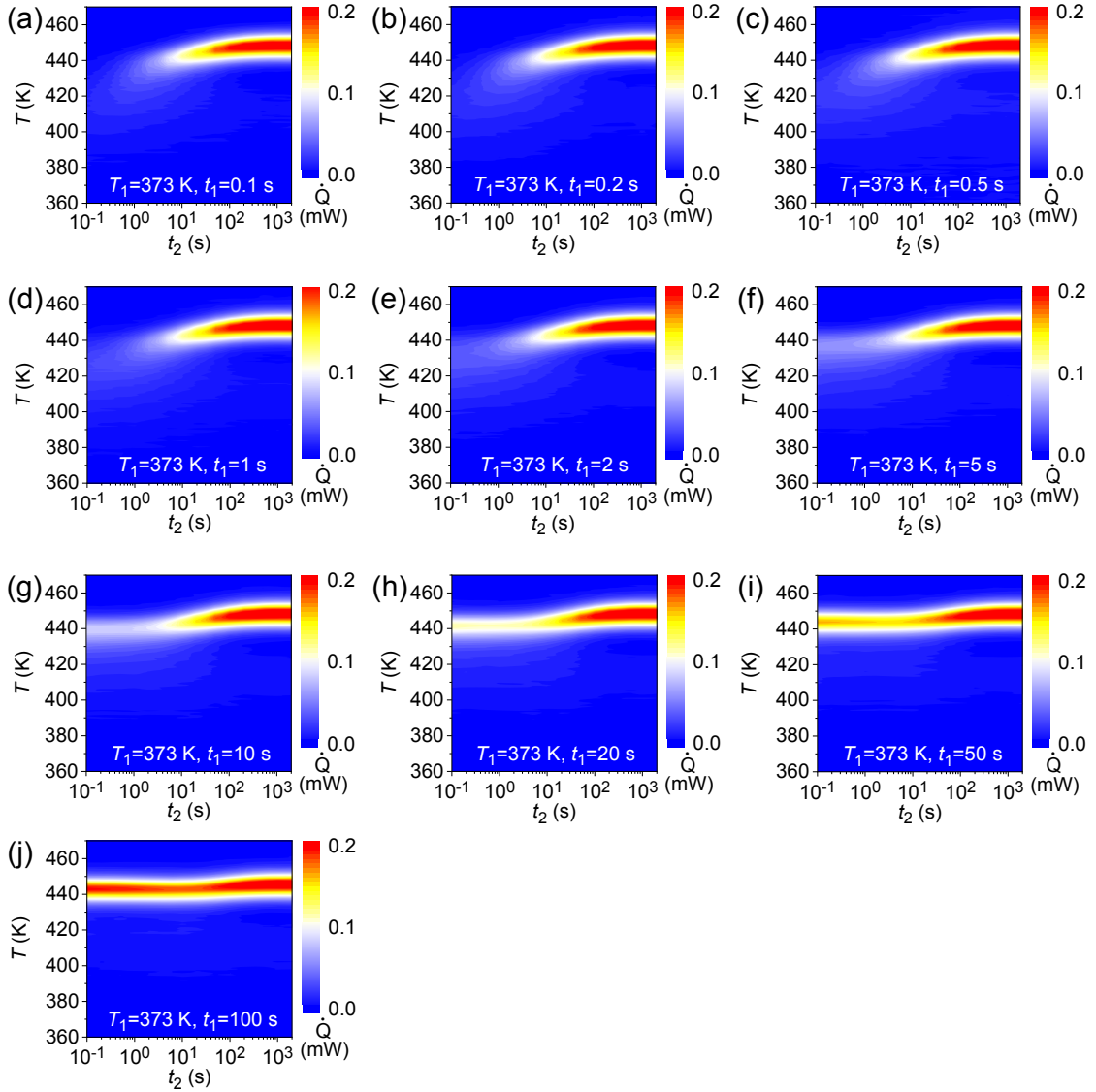


Figure S3. (a)-(j) Two-dimension contours of the relaxation heat flows of sample pre-ageing at $T_1=373$ K for $t_1=0.1$ -100 s and then ageing at $T_2=383$ K, $t_2=0.1$ -1000 s.

5. The absolute reaction rate theory

To understand the kinetic process of reaction occurs in thermal signal, the reaction peak position (T_p) versus heating rate R_h was build [1]. The form of structure defects of crystal, such as vacancy defects, can contribute to the formation enthalpy and the formation entropy. That is to say, the formation free energy $\Delta G = \Delta H - T\Delta S$. However, the metallic glasses contain lower packing and higher packing density and some results also shown that the cooperative atomic motion in relaxation and diffusion processes [2-4]. Therefore, the activation behavior of the cooperative atomic motion can not contribute to the formation enthalpy and the formation, they can contribute to the migration enthalpy and migration entropy [2]. There, we analyze the internal stress σ (units: Pa) change with time during annealing process upon an unsymmetrical barrier between basins [5]:

$$-\frac{d\sigma}{dt} \approx k'\sigma = \frac{\kappa kT}{h_p} \exp\left(-\frac{H^*}{RT}\right) \exp\left(\frac{S^*}{R}\right) \sigma \quad (1)$$

where $k' = \frac{\kappa kT}{h_p} \exp\left(-\frac{H^*}{RT}\right) \exp\left(\frac{S^*}{R}\right)$ and $\kappa = 1$ is the usual assumption [5], where k is the Boltzmann's constant, h_p is Planck's constant, R is the gas constant, H^* is migration enthalpy and S^* is migration entropy.

To describes the relaxation peak temperature change with heating rate, we consider the temperature rises at a constant heating rate R_h and $T = T_0 + R_h t$, T_0 is the initial temperature. The form of $\frac{d}{dt}\left(\frac{d\sigma}{dt}\right)$:

$$\begin{aligned} -\frac{d}{dt}\left(\frac{d\sigma}{dt}\right) &= \frac{d}{dt}\left[\frac{kT}{h_p} \exp\left(-\frac{H^*}{RT}\right) \exp\left(\frac{S^*}{R}\right) \sigma\right] \\ &= \frac{R_h}{T} \frac{kT}{h_p} \exp\left(-\frac{H^*}{RT}\right) \exp\left(\frac{S^*}{R}\right) \sigma + \frac{kT}{h_p} \exp\left(-\frac{H^*}{RT}\right) \exp\left(\frac{S^*}{R}\right) \sigma \frac{H^* R_h}{RT^2} \\ &\quad + \frac{kT}{h_p} \exp\left(-\frac{H^*}{RT}\right) \exp\left(\frac{S^*}{R}\right) \frac{d\sigma}{dt} \\ &= \frac{d\sigma}{dt} \left(-\frac{R_h}{T} - \frac{H^* R_h}{RT^2} + \frac{kT}{h_p} \exp\left(-\frac{H^*}{RT}\right) \exp\left(\frac{S^*}{R}\right)\right) \end{aligned} \quad (2)$$

The $d(d\sigma/dt)/dt = 0$ when the $d\sigma/dt$ reaches the maximum value, we have

$$\frac{R_h}{T_p} + \frac{H^* R_h}{RT_p^2} = \frac{kT_p}{h_p} \exp\left(-\frac{H^*}{RT_p}\right) \exp\left(\frac{S^*}{R}\right) \quad (3)$$

$$\frac{R_h}{T_p^2} \left(1 + \frac{H^*}{RT_p}\right) = \frac{kT_p}{h_p} \exp\left(-\frac{H^*}{RT_p}\right) \exp\left(\frac{S^*}{R}\right) \quad (4)$$

Consider the $H^*/RT_p \gg 1$, equation (4) becomes

$$\ln \frac{R_h}{T_p^3} = \ln \frac{kR}{h_p} + \ln \frac{1}{H^*} + \frac{S^*}{R} - \frac{H^*}{RT_p} \quad (5)$$

To build the relationship between the H^* and ΔE , the one order difference method was used. The differential form of the Kissinger plot [1]:

$$\Delta E = -R \frac{d \ln(\frac{R_h}{T_p^2})}{d(\frac{1}{T_p})} \quad (6)$$

The equation (6) can express as:

$$\frac{R_h}{T_p^2} = \frac{kRT_p}{h_p H^*} \exp\left(\frac{S^*}{R}\right) \exp\left(-\frac{H^*}{RT_p}\right) \quad (7)$$

and the differential form of the equation (7):

$$\begin{aligned} H^* &= -R \frac{d \ln(\frac{R_h}{T_p^2}/T_p)}{d(\frac{1}{T_p})} \\ &= -R \frac{d(\ln(\frac{R_h}{T_p^2}) + \ln(\frac{1}{T_p}))}{d(\frac{1}{T_p})} = -R \frac{d \ln(\frac{R_h}{T_p^2})}{d(\frac{1}{T_p})} - RT_p = \Delta E - RT_p \end{aligned} \quad (8)$$

That is to say,

$$\Delta E = H^* + RT_p \quad (9)$$

which is consistent with the previous result[6].

5.1 The ΔE and H^*+RT_p at four representative annealed temperatures

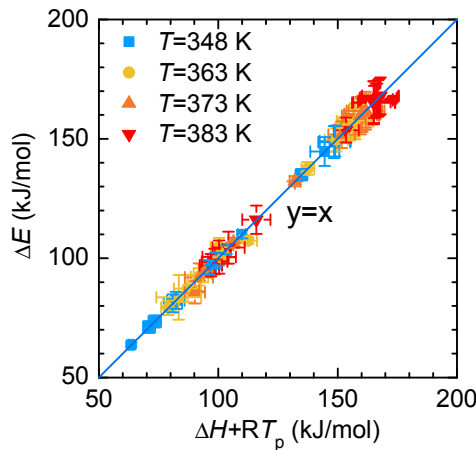


Figure S4. The ΔE versus H^*+RT_p for sample annealed at four representative annealed temperatures ($T=348$ K, 363 K, 373 K, 383 K). The solid line is the linear fit by $y=x$.

6. The enthalpy change of sample preaged at higher temperature experiments

For $T_1 > T_2$, the Δh decays monotonic with t_2 and independent of pre-ageing time $t_1=0.1, 5$ and 100 s. The memory effect dose not appear in the whole ageing temperature ($T_2=348, 363$ and 373 K).

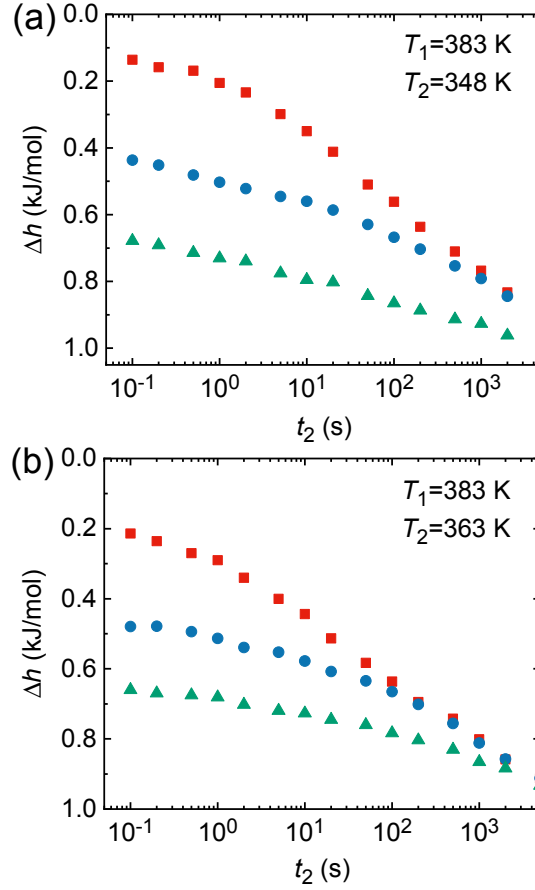


Figure S5. (a) The enthalpy change after ageing at $T_2=348$ K for various annealing time $t_2=0.1$ -5000 s, (b) The enthalpy change after ageing at $T_2=363$ K for various annealing time $t_2=0.1$ -5000 s.

7. The Tool–Narayanaswamy–Moynihan (TNM) model

7.1 The enthalpy change with time for one-step annealing condition

The Tool–Narayanaswamy–Moynihan (TNM) model has been successfully used to capture the nonexponential and nonlinear characteristics of the evolution of enthalpy and volume with annealing time. The time evolution of fictive temperature $T_f(t)$ at one-step annealing temperature T_a can be expressed [7]:

$$T_f = T_0 + (T_a - T_0) \left\{ 1 - \exp \left[- \left(\int_0^t \frac{dt}{\tau} \right)^\beta \right] \right\} \quad (10)$$

To describe the evolution of $T_f(t)$ with time, the accumulative form of equation 10 is given by:

$$T_{f,n} = T_0 + (T_a - T_0) \{ 1 - \exp [- (\sum_{i=1}^n \frac{\Delta t}{\tau_i})^\beta] \} \quad (11)$$

where T_0 is the starting point of the equilibrium state and we define $T_0=430$ K, the structure relaxation time τ_i in the form:

$$\tau_i = \exp \left\{ \ln(A) + \frac{xH^*}{RT_a} + \frac{(1-x)\Delta H^*}{RT_{f,i-1}} \right\} \quad (12)$$

where A is the constant, β is the nonexponential parameter ($0 < \beta \leq 1$), x is the nonlinear parameter ($0 < x \leq 1$), and H^* is the activation enthalpy of relaxation process and can be obtained by equation 5. Therefore, we can obtain the deviation of enthalpy change from the equilibrium state:

$$\frac{H_{tot} - H(t)}{H_{tot}} = \frac{T_f - T_a}{T_0 - T_a} = 1 - \exp \left[- \left(\int_0^t \frac{dt}{\tau} \right)^\beta \right] \quad (13)$$

7.2 The enthalpy change with time for two-step annealing condition

The time evolution of T_f for two step annealing conditions can be given by the superposition of the responses of equation 10, which is expressed as [7]:

$$T_f =$$

$$T_0 + (T_1 - T_0) \left\{ 1 - \exp \left[- \left(\int_{t'}^{t''} \frac{dt}{\tau} + \int_{t_2}^t \frac{dt}{\tau} \right)^\beta \right] \right\} + (T_2 - T_1) \left\{ 1 - \exp \left[- \left(\int_{t''}^t \frac{dt}{\tau} \right)^\beta \right] \right\} \quad (14)$$

the accumulative form of equation 14 is given by:

$$T_{f,n} = T_0 + (T_1 - T_0) \left\{ 1 - \exp \left[- \left(\sum_{i=1}^{n_1} \frac{\Delta t}{\tau_i} + \sum_{j=n_1}^n \frac{\Delta t}{\tau_j} \right)^\beta \right] \right\} + (T_2 - T_1) \left\{ 1 - \exp \left[- \left(\sum_{j=n_1}^n \frac{\Delta t}{\tau_j} \right)^\beta \right] \right\} \quad (15)$$

where T_1 and T_2 are the first annealing temperature and the second annealing temperature, respectively. The structure relaxation time τ_i corresponds to the T_1 and τ_j corresponds to the T_2 , they are given by:

$$\tau_i = \exp \left\{ \ln(A) + \frac{xH^*}{RT_1} + \frac{(1-x)H^*}{RT_{f,i-1}} \right\} \quad (16)$$

$$\tau_j = \exp \left\{ \ln(A) + \frac{xH^*}{RT_2} + \frac{(1-x)H^*}{RT_{f,j-1}} \right\} \quad (17)$$

Therefore, we can also have the form of enthalpy change with time at two step annealing conditions:

$$\frac{H_{tot} - H(t)}{H_{tot}} = \frac{T_f - T_2}{T_0 - T_2} \quad (18)$$

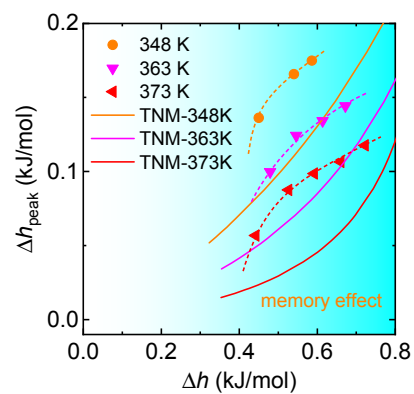


Figure S6. The strength of memory effect (Δh_{peak}) versus enthalpy evolution (Δh) during the first annealing step at $T=348, 363, 373$ K. Δh_{peak} is obtained by subtracting the initial enthalpy from the peak enthalpy. The memory effect appears only for Δh greater than about 0.4 kJ/mol. The solid symbols represent experimental data in which the memory effect was observed. The dashed lines are for guides for the eyes. The solid curves are fitting results using TNM model.

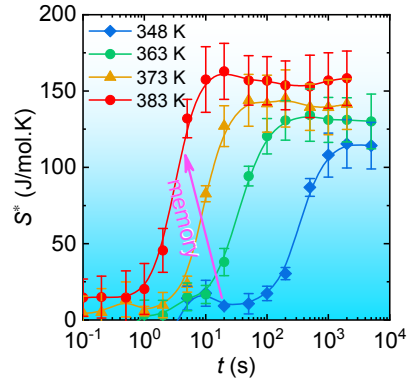


Figure S7. S^* as a function of annealing time for single-step annealing at $T=348, 363, 373, 383$ K.

The arrow shows that the memory effect appears only if the glass jumps from low temperature (with a small S^*) to high temperature (with a big S^*). As the enthalpy relaxation exhibits strong dependence on temperature, the up-jump in the two-step memory experiments

References

- [1] H. E. Kissinger, *Reaction kinetics in differential thermal analysis*, Anal. Chem. **29**, 1702 (1957).
- [2] U. Geyer, S. Schneider, W. L. Johnson, Y. Qiu, T. A. Tombrello, and M. Macht, *Atomic diffusion in the supercooled liquid and glassy states of the $Zr_{41.2}Ti_{13.8}Cu_{12.5}Ni_{10}Be_{22.5}$ alloy*, Phys. Rev. Lett. **75**, 2364 (1995).
- [3] F. Faupel, P. W. Huppe, and K. Ratzke, *Pressure dependence and isotope effect of self-diffusion in a metallic glass*, Phys. Rev. Lett. **65**, 1219 (1990).
- [4] Y. Fan, T. Iwashita, and T. Egami, *How thermally activated deformation starts in metallic glass*, Nat. Commun. **5**, 5083 (2014).
- [5] F. W. Cagle and H. Eyring, *An application of the absolute reaction rate theory to some problems in annealing*, J. Appl. Phys. **22**, 771 (1951).
- [6] H. W. Starkweather, *Simple and complex relaxations*, Macromolecules **182**, 1277 (1981).
- [7] C. T. Moynihan *et al.*, *Structural relaxation in vitreous materials*, Ann. N. Y. Acad. Sci. **279**, 15 (1976).